

Engagement, Ohio

A Coordinated Visual Analytics Tool for Challenge 1

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The Challenge

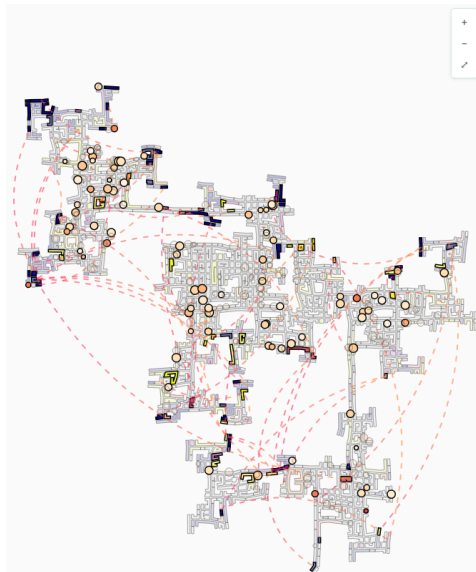
VAST Mini-Challenge 1 — describe a town from log data.

Data

- 1,011 participants, 1,042 buildings
- 253 employers, 5 schools
- 7.5M social interactions
- ~5 GB activity logs (MC2 — out of scope)

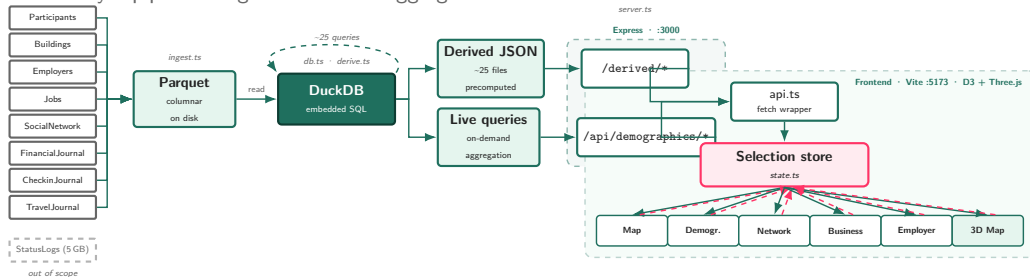
Three questions

- Q1 — Demographics
- Q2 — Social network patterns
- Q3 — Business base



Architecture

Three-layer pipeline: ingest & store · aggregate & serve · render & coordinate



—→ data & fetch flow - - -> selection write-back

Architecture — Why These Choices

Backend

- **Parquet over CSV:** columnar I/O is 10× faster for the aggregation queries we actually run.
- **DuckDB over pandas:** SQL is more reproducible, embedded means no server to manage. Full derive runs in ~60s.
- **Two API paths:** *precomputed JSON* for everything the frontend needs at startup, *live SQL endpoints* for the handful of queries that depend on user context (e.g. demographics breakdown of a selection).

Frontend

- **One selection store, six views:** all views subscribe to the same state; any view can write to it. Coordination is structural, not retrofitted.
- **Source-tagged selections:** every selection carries its origin (chart, building, employer, region). Views can render differently for the source vs. secondary highlights.
- **D3 + Three.js:** D3 for SVG composability and joinable enter/exit transitions; Three.js with a custom GLSL shader for the animated 3D arcs.

Design Decision 1 — Scope Discipline

We deliberately did not ingest the 5 GB ParticipantStatusLogs.

Why? Those are 5-minute snapshots of every participant's location and balance — *Patterns of Life* data, which belongs to MC2.

What we gained

- Clean MC1/MC2 boundary in our analysis
- Sub-minute derive pipeline
- Reproducible repo <80 MB

What we acknowledged

- 131 participants drop out after month 1
- This survivorship bias is flagged throughout the report

The honest framing

“We answer Q1, Q2, Q3 from structural data alone. Temporal behaviour (when people eat, work, socialise) belongs to MC2 and is out of scope.”

Reviewers reward this kind of explicit scope statement.

Design Decision 2 — Coordination over Tabs

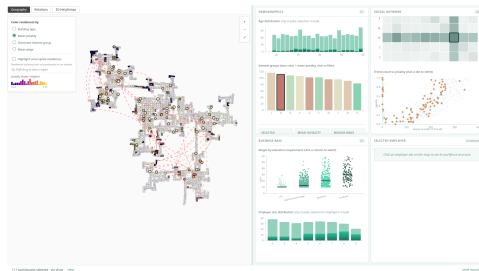
Single shared selection across map + 4 dashboard panels.

Click a building, click an interest group, lasso a region, click an employer — the *same* selection drives every view.

Why this matters

- Comparative exploration without tab-switching
- All four panels stay visible — your eyes find the correlations, not your clicks
- Most VAST tools use tabbed dashboards. Ours is deliberately Tableau-style 2×2 grid.

Cost: minimum viewport 1280×800.



One click on a Group H bar — map, demographics, network, and business panels

all update.

Design Decision 3 — Colour Choices Have Consequences

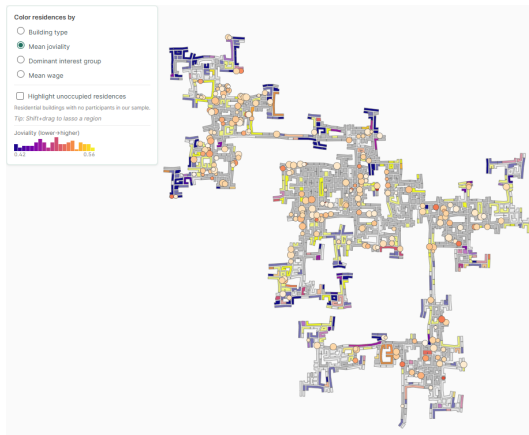
Joviality was originally rendered with RdBu (red-blue divergent).

Reviewers (and our supervisor) noticed:

- White midpoint $\approx$ “no data” on a map
- 0.5 has no semantic meaning for joviality
- Most of the data lives in $[0.42, 0.56]$ — the *theoretical* range washed out the actual variation

We switched to plasma on the empirical domain.

- Sequential, perceptually uniform
- No semantic midpoint problem
- Uses the full colormap range for actual data



Plasma joviality on $[0.42, 0.56]$ — what the data actually does.

Three Findings — A Teaser for the Demo

Workplaces are social hubs

65% of coworker pairs are friends.
Baseline (random pairs): 16%.
Top 10 employers: 100% pair cohesion.

The Group H paradox

Largest group (111).
Highest interaction count.
Lowest joviality.
Activity $\neq$ wellbeing.

Friends predict happiness

<50 friends $\rightarrow$ joviality <0.3
 >200 friends $\rightarrow$ joviality >0.6
The strongest correlate in the dataset.

More in the demo.

Live Demo

~4 minutes

- Q1 Demographics overview, vacancy highlights
- Q2 Group H paradox, friendship arcs across the city
- Q3 Employer selection, workplace cohesion
- 3D Heightmap with animated arcs

Lessons Learned

What worked

- Strict scope discipline saved us weeks
- DuckDB + Parquet pipeline is far faster than pandas
- Cross-view coordination earns its complexity
- Honest methodological framing (vanishing cohort, confidence threshold) reads as mature

What we'd do differently

- Settle on a colour palette earlier
- Build the selection store *before* the views
- More time on the report, less on adding features
- Performance optimisation (3D arc instancing) deferred to “next iteration”

Team Organisation

- **Backend pipeline** (DuckDB, SQL aggregation, Parquet I/O, REST endpoints, derive script) — Reda
- **Frontend views** (map, demographics, social network, business base, employer detail, 3D heightmap) — Diogo, Leonint, Jerry, Reda
- **Q1 analysis and prose** — Jerry
- **Q2 analysis (10 patterns)** — Diogo
- **Statistical group analysis** (Kruskal-Wallis, eta-squared) — Leonint
- **Report integration and figures** — Reda

Total effort: ~120 hours across 5 weeks.

Repo: github.com/Redabelcaid/HPCDA-Project-Demographics-Relationships

Questions

5 minutes

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Leonint Amarantos

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